

Real-Time Applications with Learning Outcomes

1. Chat Application

Core Outcomes

- **WebSocket Mastery:** Implement real-time bidirectional communication
- **Message Persistence:** Store and retrieve chat history with pagination
- **Typing Indicators:** Show when users are typing in real-time
- **File Sharing:** Enable image/document sharing within chats
- **Group Chats & Channels:** Create public/private chat rooms with moderation
- **Read Receipts:** Display message delivery and read status

Technical Skills Gained

- **Socket.io/WebSocket:** Real-time event handling
- **Message Queueing:** Handle high-volume message delivery
- **Presence System:** Track online/offline status
- **Media Compression:** Optimize file uploads for chat
- **Security:** End-to-end encryption implementation

2. Live Polling/Voting System

Core Outcomes

- **Real-time Results:** Instant vote tally visualization
- **Live Updates:** See votes as they happen without refresh
- **Poll Management:** Create, edit, and moderate polls
- **Audience Participation:** QR code joining, anonymous voting
- **Results Export:** Download voting data for analysis

Technical Skills Gained

- **Data Synchronization:** Keep all clients in sync with live data
- **Chart Animation:** Animate vote updates in real-time
- **Rate Limiting:** Prevent voting abuse and spam
- **Event Broadcasting:** Efficiently broadcast to large audiences
- **Analytics:** Track participation and engagement metrics

3. Collaborative Drawing/Whiteboard

Core Outcomes

- **Multi-user Canvas:** Simultaneous drawing by multiple users
- **Drawing Tools:** Implement brush, shapes, text, and color tools
- **Session Management:** Save and load drawing sessions

- Cursor Tracking: See other users' cursors in real-time
- Version History: Undo/redo with collaborative awareness

Technical Skills Gained

- Canvas API: HTML5 Canvas manipulation
- Real-time Sync: Conflict resolution for overlapping edits
- Data Optimization: Efficient transmission of drawing data
- Tool State Management: Sync tool states across users
- Export Formats: Save as PNG, SVG, or PDF

4. Notification System

Core Outcomes

- Multi-channel Delivery: Push, email, SMS, and in-app notifications
- Smart Routing: Deliver to appropriate channels based on user preference
- Notification Center: Unified inbox for all notification types
- Delivery Analytics: Track open rates and engagement
- Scheduling: Send notifications at optimal times

Technical Skills Gained

- Service Workers: Implement push notifications
- Queue Systems: Handle notification delivery at scale
- Template Management: Create reusable notification templates
- A/B Testing: Test notification effectiveness
- GDPR Compliance: Manage user consent and preferences

5. Stock Price Ticker

Core Outcomes

- Real-time Data: Live stock price updates from financial APIs
- Portfolio Tracking: User investment portfolio simulation
- Price Alerts: Custom alerts for price thresholds
- Historical Data: Interactive charts with historical performance
- News Integration: Related financial news alongside prices

Technical Skills Gained

- WebSocket Streaming: Handle high-frequency data updates
- Financial APIs: Integrate with Alpha Vantage, Yahoo Finance, etc.
- Data Visualization: Implement interactive financial charts
- Caching Strategy: Optimize API calls for rate limits
- Web Workers: Offload heavy calculations from main thread